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A PROJECT REPORT

ON

GuardianEye: Multimodal Late Fusion of Image and Audio Poaching Indicators

Submitted in partial fulfillment of the requirements for the degree of

Bachelor of Technology

in

Artificial Intelligence and Machine Learning

Subject: Multimodal AI (MAI)

Group 27

Academic Year: 2026–2027

Submitted by

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CERTIFICATE

This is to certify that the project work entitled “GuardianEye: Multimodal Late Fusion of Image and Audio Poaching Indicators” is carried out by the students named below, of the Department of Artificial Intelligence and Machine Learning, Symbiosis Institute of Technology, Pune, in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Artificial Intelligence and Machine Learning of Symbiosis International (Deemed University), Pune, during the academic year 2026–2027.

The work is submitted for the course Multimodal AI (MAI), Group 27, and was completed under the guidance of Dr. Nivedita Mishra.

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DECLARATION

We hereby declare that the project report entitled “GuardianEye: Multimodal Late Fusion of Image and Audio Poaching Indicators” is a record of work carried out by us at the Department of Artificial Intelligence and Machine Learning, Symbiosis Institute of Technology, Pune, a constituent of Symbiosis International (Deemed University), under the guidance of Dr. Nivedita Mishra, during the academic year 2026–2027.

This report is submitted in partial fulfillment of the requirements for the degree of Bachelor of Technology in Artificial Intelligence and Machine Learning, for the course Multimodal AI (MAI), Group 27.

We further declare that this report has not been submitted to any other university or institute for the award of any degree or diploma.

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Certified that the above statement made by the students is correct to the best of our knowledge and belief.

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ACKNOWLEDGEMENT

The authors acknowledge Dr. Nivedita Mishra, Assistant Professor, Department of Artificial Intelligence and Machine Learning, Symbiosis Institute of Technology, for guidance of this Multimodal AI work during the academic year 2026–2027.

The authors acknowledge Dr. Sumit Kumar, Head of the Department of Artificial Intelligence and Machine Learning, Symbiosis Institute of Technology, Pune, and Symbiosis International (Deemed University), for the academic framework in which the project was undertaken.

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ABSTRACT

GuardianEye is a multimodal review pipeline for wildlife evidence. Two sensory streams are modelled separately and combined only at decision time. Still frames from camera traps and from aerial thermal imagery pass through a YOLOv8 detector trained independently on each visual domain. Short audio clips are converted to log-mel spectrograms and classified by a convolutional network. Video is not treated as a third modality; frames are sampled and routed through the image branch. A human box, a vehicle box, or a gunshot-like score is an indicator. One detection is not proof of poaching.

The multimodal design is late fusion. Each branch keeps its own representation, loss, and evaluation protocol. Image metrics are written separately for aerial_thermal and camera_trap_rgb. Audio is split by site: Belize to Vietnam, Vietnam to Belize, and an in-domain control for each site. Empty frames and background clips remain in the prepared sets. Rare visual classes are oversampled. Audio training uses class-weighted loss. Because no public collection used here aligns an image with the audio of the same moment, fusion pairs are constructed from labels and marked as a synthetic proxy protocol.

On the corpus present in this working tree the prepared set contains 22 images, 29 audio windows, and 16 synthetic proxy pairs. Detection scores are written only by the evaluation scripts, and only when a checkpoint is present. The model directory contains no checkpoint. The FastAPI service and Gradio dashboard expose multimodal inference with honest model-info reporting when weights are absent.

Keywords: multimodal AI, late fusion, YOLOv8, log-mel CNN, site stratification, synthetic proxy protocol, missing-modality degradation, audit log.

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1. Introduction

Patrol records arrive as heterogeneous sensory streams: RGB camera-trap stills, aerial thermal frames, and short acoustic clips. The streams are rarely synchronised. A single sensor reading is incomplete, and a single detector output is ambiguous. GuardianEye addresses this as a multimodal AI problem. Separate unimodal models extract modality-specific indicators. A late-fusion stage combines those indicators into a risk level that a ranger can review.

Video is not a third modality. Frames are sampled from it and submitted to the image branch. The image branch itself spans two domains: aerial_thermal and camera_trap_rgb. A detector trained on one domain is not substituted for the other. The audio branch is stratified by recording site so that geographic transfer can be measured rather than hidden inside a shuffled pool.

There is no labelled poacher class in the collections named by the configuration, and there is no time-aligned field incident that pairs an image with the audio of the same moment. The system therefore detects indicators and fuses them. It does not claim a confirmed poaching event. Configuration lives in configs/*.yaml. The seed is 27. Download URLs remain TODO until a person fills them in.

2. Multimodal Problem Formulation

2.1 Modalities and indicator classes

The visual modality carries four outcomes: human, vehicle, animal, and empty. An empty image is a frame with no box. It is a negative example, and it stays in the training set. Visual poaching indicators are restricted to human and vehicle. Animal boxes are recorded and are not treated as poaching evidence.

The acoustic modality carries two labels: background and gunshot. The gunshot class is a gunshot-like impulse. Background covers ambient sound and impulse-like distractors tagged on the manifest. Background clips are retained so that the classifier sees the negative class in proportion to the field mixture.

2.2 Why late fusion

Early fusion would require a shared representation of image and audio before either branch is trained. The repositories used here do not supply paired, time-aligned samples at scale. Late fusion keeps each encoder specialised, permits independent calibration, and degrades gracefully when one modality is missing. The fusion stage reads detection scores and timestamps, not raw pixels or waveforms.

2.3 Decision policy

The primary policy is rules mode in configs/fusion.yaml. Inclusive thresholds of 0.50 mark a strong image indicator and a strong gunshot score. When only one reliable modality is present, unimodal thresholds of 0.85 apply, and the level cannot be HIGH. Coincidence uses a 5.0 second expansion of the audio window.

HIGH requires a human or vehicle detection and a gunshot score, both at or above threshold, inside that window. MEDIUM covers exactly one strong indicator, or both indicators outside the window. LOW covers neither. Alerting is configured for HIGH only. Weight profiles balanced, night, and wind rescale the numeric score when weighted mode is selected; rules mode remains the documented primary path.

3. System Architecture

The pipeline is organised as two parallel unimodal branches and one fusion module. The image branch runs YOLOv8 once per visual domain. The audio branch converts each clip to a fixed log-mel matrix and classifies it with a small convolutional network. The fusion module implements the rule set above, writes an audit row to SQLite, and exposes the result through FastAPI and a Gradio dashboard at /dashboard.

Missing-modality handling is explicit. An upload that contains only an image or only an audio clip still receives a score. The response states which branch was reliable. GET /model-info reports, for each domain and for audio, whether a checkpoint was loaded. When weights are absent, the service states that the corresponding evidence is not reliable.

4. Visual Modality

The image trainer runs one YOLOv8 model per domain. configs/image.yaml sets train_per_domain to true and names yolov8n.pt as the base weights. The full schedule is 50 epochs, image size 640, batch size 8, and device cpu by default. Aerial thermal frames use a replicate three-channel representation so that an RGB backbone can be reused without inventing false colour structure.

Class imbalance on the image side is handled by oversampling, not by dropping empty frames. Rare box classes are duplicated in a copy of the training split until their instance counts meet the most common non-empty class under the configured multiplier. Evaluation is per domain. guardianeye.eval.image_eval scores aerial_thermal and camera_trap_rgb separately. When weights exist it can write detection metrics to reports/. Those numbers are never hardcoded in this report.

5. Audio Modality and Cross-Site Transfer

Audio preparation resamples to 16 kHz and cuts windows of 3.0 seconds. Each window becomes a log-mel matrix of 64 bins. The default classifier is a three-block convolutional network with batch normalisation and ReLU activations, followed by adaptive pooling and a linear head over the two audio classes. Training uses class-weighted cross-entropy so that the rarer gunshot class retains gradient mass.

Four experiments isolate geographic transfer. belize_to_vietnam trains and validates on Belize and tests on all of Vietnam. vietnam_to_belize reverses the sites. in_domain_belize and in_domain_vietnam each split one site into train, validation, and test. Validation is drawn only from the training site. This design is a multimodal evaluation discipline: the acoustic encoder is judged under domain shift that the fusion stage will later inherit.

6. Late Fusion and Synthetic Proxy Evaluation

No public set used by this repository aligns an image with the audio of the same incident. scripts/build_paired_events.py therefore constructs pairs from labels. Every row carries protocol synthetic_proxy. The construction is a proxy for multimodal coincidence, not a substitute for field-aligned capture.

Eight scenarios are applied in each image domain: human with gunshot, vehicle with gunshot, small-human with gunshot, human with background, vehicle with background, animal with background, empty with gunshot, and empty with an impulse distractor. Expected levels on the proxy set follow the fusion rules. Fusion evaluation, when both branches can be loaded, compares image-only, audio-

only, and fused decisions and writes reports/fusion/. A missing checkpoint leaves that report empty of scores.

7. Multimodal Serving Interface

The service is a FastAPI application bound to port 8742. A Gradio dashboard is mounted at /dashboard and accepts image and audio uploads in one interaction. GET /health reports that the process is up. The audit store is SQLite at data/events.sqlite. Snapshots and audio copies are written under data/events/. Every decision is logged, including events that do not alert.

The dashboard banner states that the system is a proof-of-concept for indicators, not confirmed poaching. Model-info text surfaces unreliable weights when checkpoints are absent, so a multimodal demo cannot be mistaken for a calibrated field detector.

8. Empirical Characterisation of the Multimodal Corpus

The figures below describe the prepared multimodal corpus in this working tree. They are metadata counts. They are not precision, recall, F1, mAP, or false-positive rate. Those detection metrics are produced only by evaluation scripts after weights exist. The models/ directory contains no checkpoint.

8.1 Images

The image manifest contains 22 records, 11 in aerial_thermal and 11 in camera_trap_rgb. 12 empty images are retained. Imbalance ratios are 6.0 and 6.0.

Table 1. Image records by domain and primary class.

Domain	Human	Vehicle	Animal	Empty	Total	Imbalance
aerial_thermal	2	1	2	6	11	6.0
camera_trap_rgb	2	1	2	6	11	6.0

Table 2. Image records by domain and split.

Domain	Train	Validation	Test
aerial_thermal	7	3	1
camera_trap_rgb	7	3	1

8.2 Audio

The audio manifest contains 29 windows. Belize contributes 14; Vietnam contributes 15. 20 background clips are retained.

Table 3. Audio windows by site and label.

Site	Gunshot	Background	Total	Imbalance
Belize	4	10	14	2.5
Vietnam	5	10	15	2.0

Table 4. Audio split sizes by experiment.

Experiment	Train	Validation	Test

belize_to_vietnam	11	3	14
in_domain_belize	8	3	3
in_domain_vietnam	8	3	3
vietnam_to_belize	11	3	14

8.3 Synthetic proxy pairs

The pair file contains 16 rows. All carry protocol synthetic_proxy. Expected levels on the proxy set are HIGH 6, MEDIUM 6, and LOW 4.

Table 5. Synthetic proxy pairs by scenario.

Scenario	Rows	Role in the proxy protocol
animal_background	2	Expected LOW
empty_gunshot	2	Expected MEDIUM
empty_impulse	2	Expected LOW
human_background	2	Expected MEDIUM
human_gunshot	2	Expected HIGH
small_human_gunshot	2	Expected HIGH
vehicle_background	2	Expected MEDIUM
vehicle_gunshot	2	Expected HIGH

Table 6. Fusion decision levels on the proxy set (expected).

Expected level	Pairs
HIGH	6
MEDIUM	6
LOW	4

8.4 Serving observation

The audit log in this working tree contains 1 event. Alerted 0; not alerted 1. The recorded reason includes unreliable_weights when checkpoints are absent. That outcome is consistent with a multimodal service that refuses to overstate evidence from unloaded branches.

Empty images and background audio dominate the prepared counts and were retained. The multimodal protocol therefore trains and evaluates against the negative class rather than discarding it. No detection metric is stated here because no checkpoint is present to score.

References

[1] GuardianEye configuration files configs/data.yaml, configs/image.yaml, configs/audio.yaml, configs/fusion.yaml, and configs/serve.yaml. Repository working tree. Academic year 2026–2027. Download URLs left as TODO.

[2] Ultralytics YOLOv8. Base weights named yolov8n.pt in configs/image.yaml. One training run is defined per image domain.

[3] Named source collections in configs/data.yaml, with URLs left as TODO: BIRDSAI (aerial thermal); Caltech Camera Traps, including empty frames; Snapshot Serengeti; the Belize tropical-forest gunshot set; the Chu Mom Ray Vietnam set.

[4] FastAPI service on port 8742 with Gradio dashboard at /dashboard. SQLite audit table events at data/events.sqlite.

[5] Prepared multimodal metadata summarised in reports/bda/summary.json (corpus counts only; not model metrics).